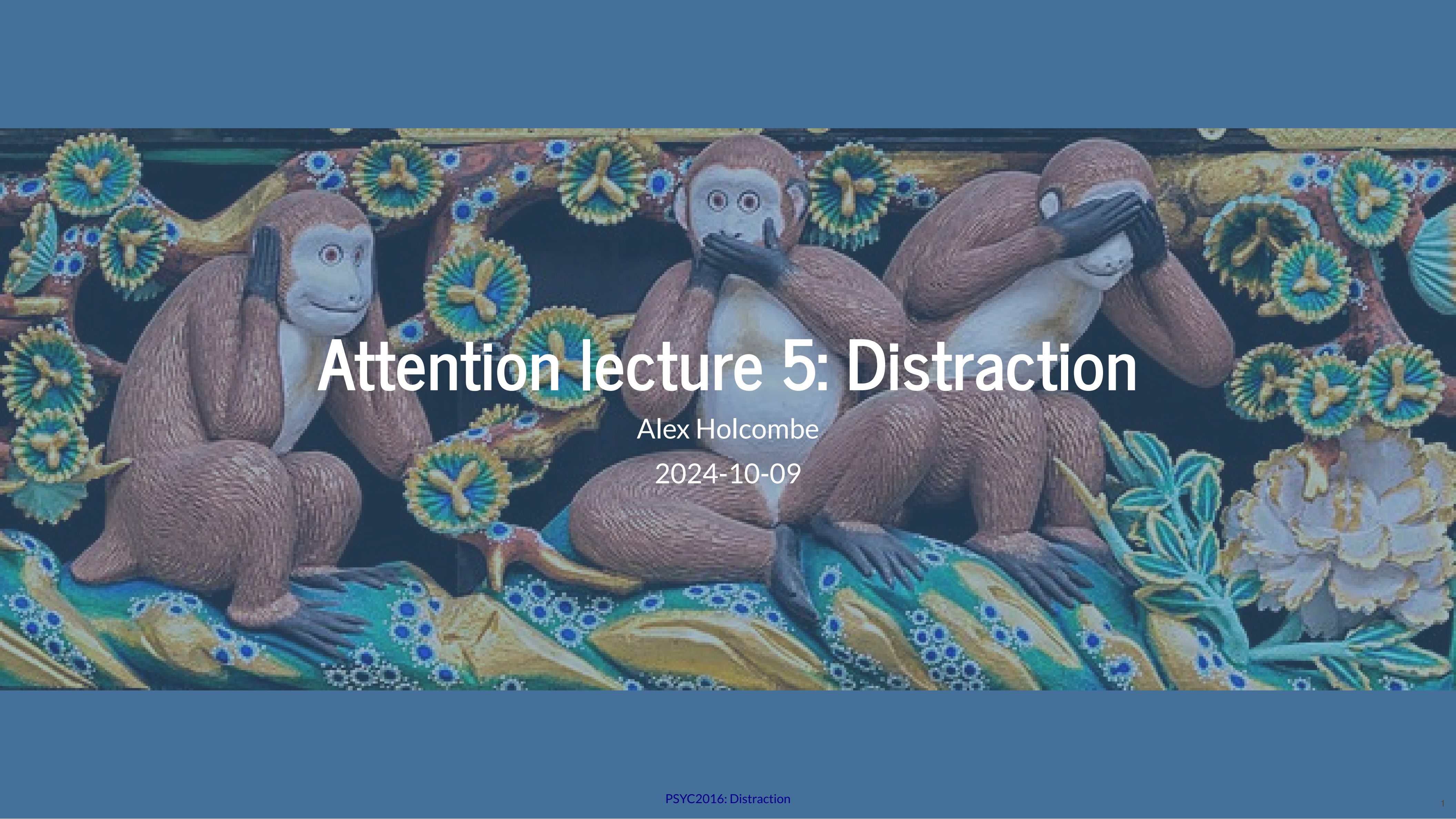




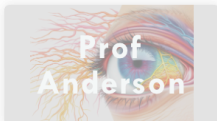
Attention lecture 5: Distraction

Alex Holcombe

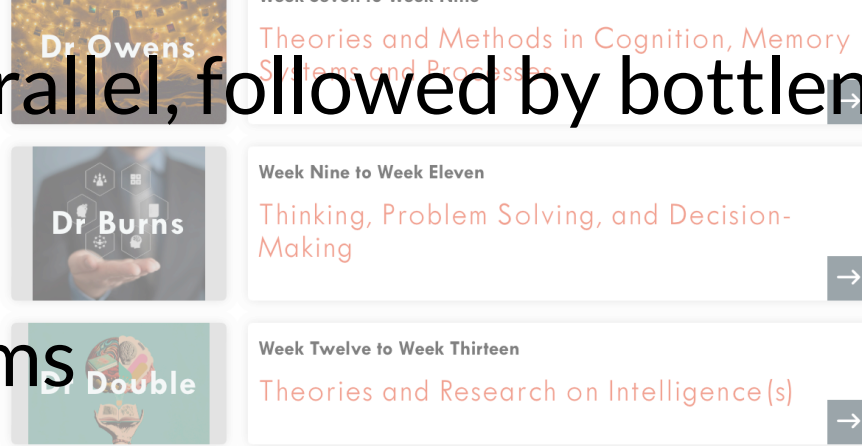
2024-10-09

Attention: a bridge between Perception and Cognition



	Week One to Week Four Vision	→
	Week Four to Week Five Hearing & Consciousness	→
	Week Six to Week Seven Attention PSYC2016: Distraction	→
	Week Seven to Week Nine	

- Our mental architecture is massively-parallel, followed by bottleneck(s)
- To deal with this architecture, we have:
 - Crude salience and alerting mechanisms
 - Guidance of attention by memory
- This helps explain distractions



Outline of Lectures 1-4

1. Why we need selective attention

- Overt attention
- Bottleneck architecture of the visual system necessitates covert attention
- Need for vigilance
 - Because of bottleneck, we evolved crude salience and alerting mechanisms

2. and 3. Kinds of selection, parallel versus serial processing

- Bottom-up attention (Chapters 7 and 9)
- Our behavior reflects competition between top-down and bottom-up
- Combinations of features typically aren't processed in parallel
 - Selective attention is needed to combine features (Treisman)
- Feature selection is compulsorily global
 - Maybe because What processing (features) occurs on different pathway than Where pathway, which mediates spatial attention

4. Shape

- Combining features within a feature dimension
- Quantifying shape processing efficiency with dipoles

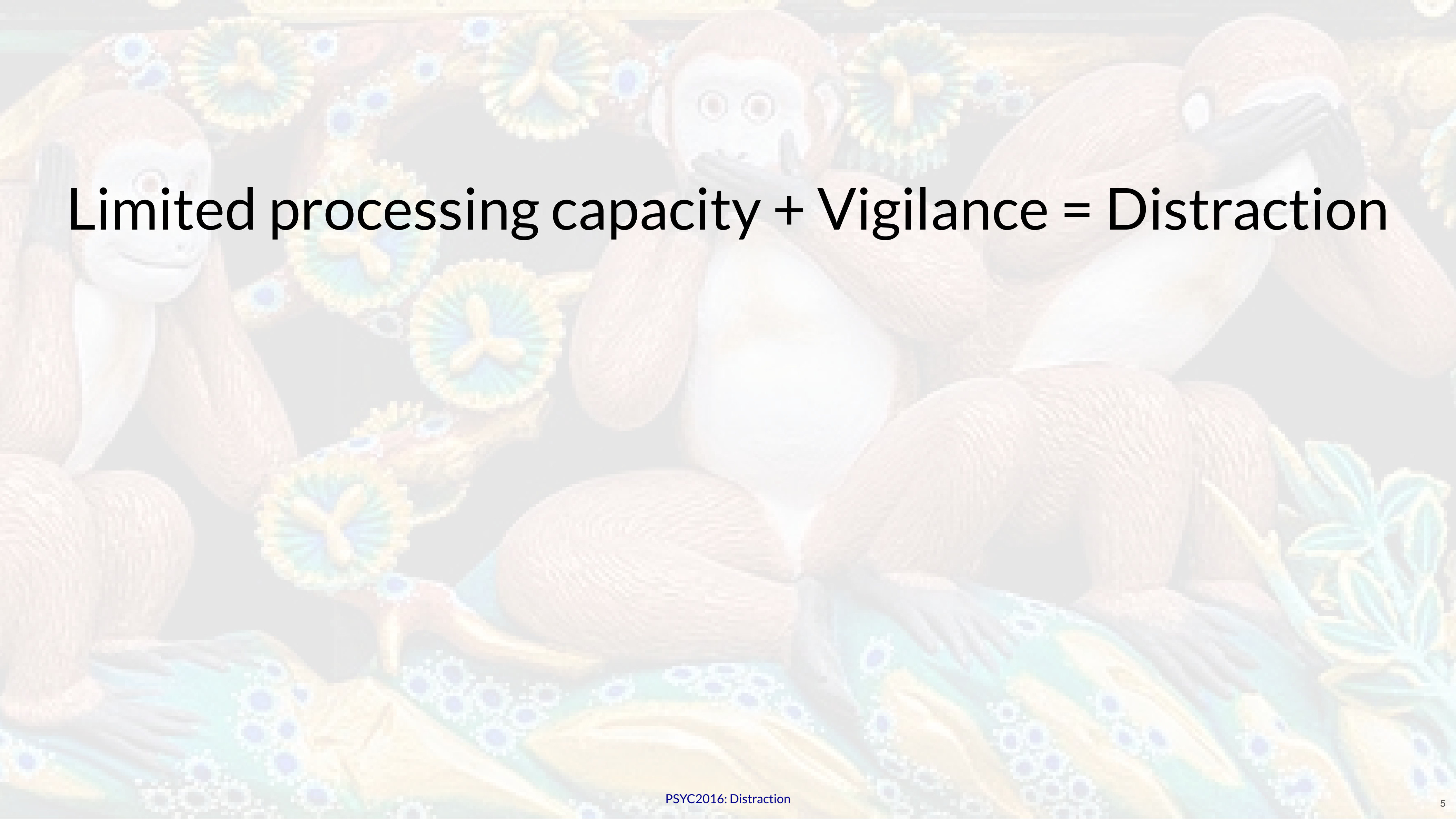
4. Change blindness

- In blank screen sandwiches, have to attend to location to detect change
- A “grand illusion” of visual experience?
- Poor peripheral vision can prevent change detection
- Role of flicker / motion detectors
- Memory for important locations can drive attention

2,3. Kinds of selection, parallel versus serial processing

Lecture 5: Distraction





Limited processing capacity + Vigilance = Distraction

A blurred background image showing a group of students in a lecture hall. In the foreground, a student with light-colored hair is looking down at a white piece of paper. Other students are visible in the background, some looking towards the front of the room. The overall scene is out of focus, emphasizing the text in the foreground.

What most distracts you during lecture?

Phones!

-



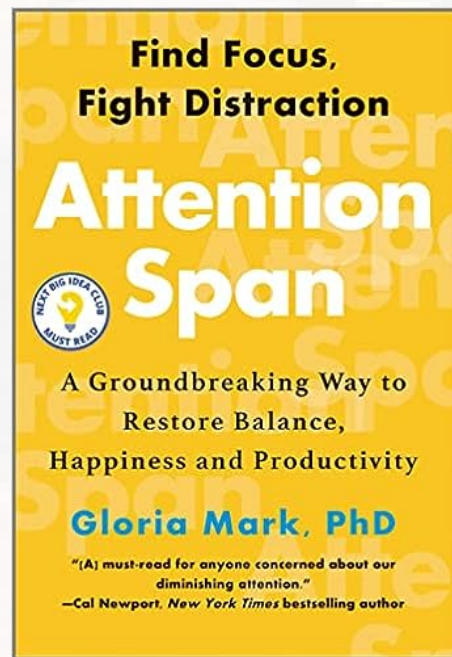
Ezra Klein on mobile phones



- In 1990, imagine I told you that I was going to invent a sinister tool...
- As people used it, their attention spans would degrade, as the tool would constantly shift their focus, weakening their powers of concentration and contemplation.
- This tool would show people whatever it is they found most difficult to look away from — which would often be what was most threatening about the world, from the worst ideas of their political opponents to the deep injustices of their society.
- It would fit in their pockets and glow on their night stands and never truly be quiet; there would never be a moment when people could be free of the sense that the pile of messages and warnings and tasks needed to be checked.
- What would you have thought this engine of distraction, division, and cognitive fracture would do to humanity?

Phones!





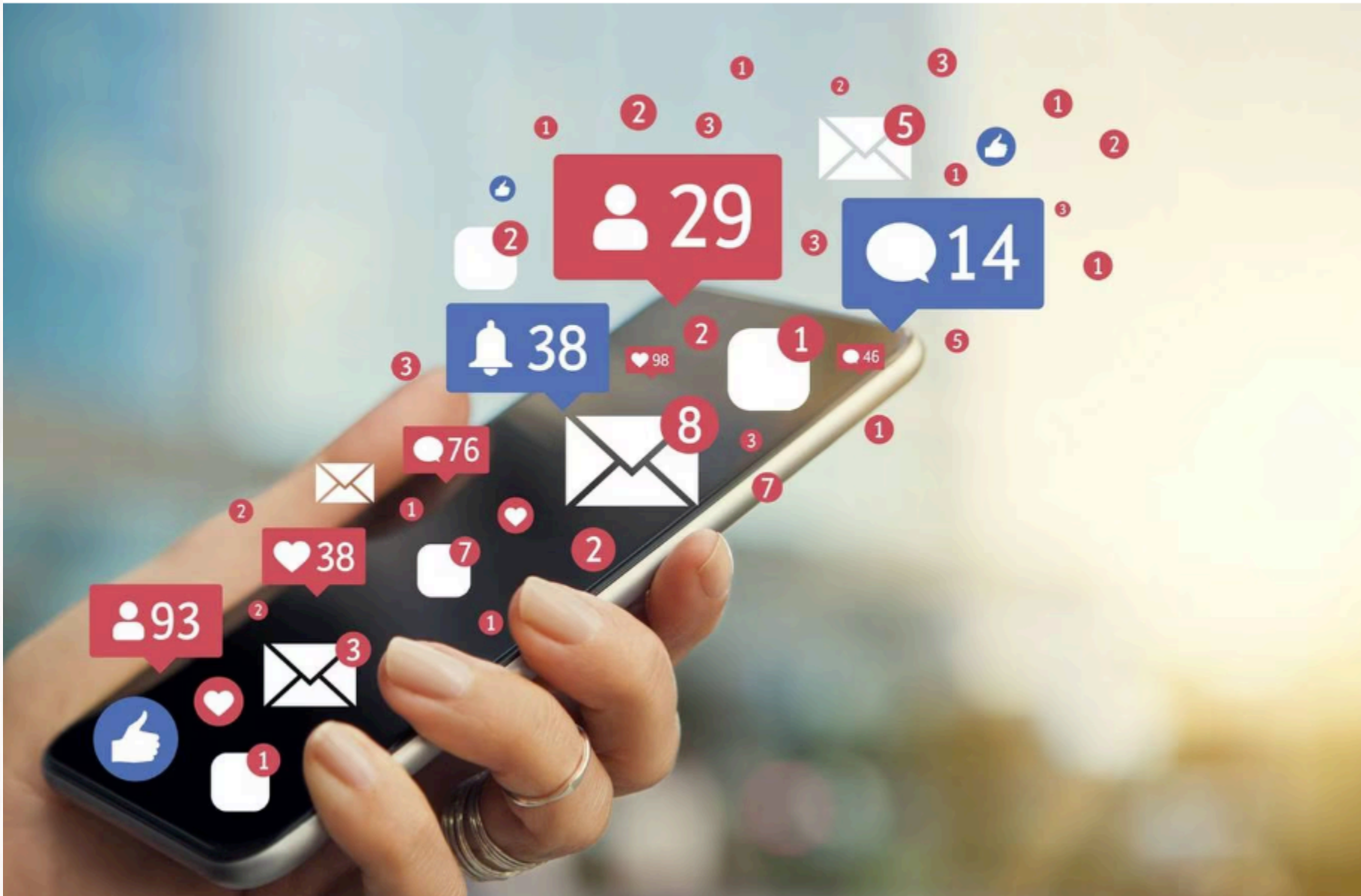
Our mind becomes like a pinball propelled from lever to lever by text chimes, social media notifications and targeted ads...

Without top-down control of our attention, we open ourselves up to stimuli that steer our attention for us.

Controlling bottom-up sensory events

Notification detox

One ping after another: Why we all need a notification detox
They alert us to calls, texts, even exercise times. Should we turn them all off for good?



Irish Times [article](#)

Take some advice from the experts on phones and apps!

The New York Times

A Silicon Valley School That Doesn't Compute

 Share full article





 +

A photograph of a woman, Bryn Perry, lying on her back on a wooden school desk. She is wearing a light blue long-sleeved shirt and dark pants, and is holding an open book above her head, reading. The desk has two nameplates: one on the left that says 'Cristina' and one on the right that says 'KENZIE'. In the background, there is a wooden cabinet with several framed pictures and a potted plant.

The Waldorf School in Los Altos, Calif., eschews technology. Here, Bryn Perry reads on a desktop. Jim Wilson/The New York Times

By Matt Richtel
Oct. 22, 2011

LOS ALTOS, Calif. — The chief technology officer of eBay sends his children to a nine-classroom school here. So do employees of Silicon Valley giants like Google, Apple, Yahoo and Hewlett-Packard.

But the school's chief teaching tools are anything but high-tech: pens and paper, knitting needles and, occasionally, mud. Not a computer to be found. No screens at all. They are not allowed in the classroom, and the school even frowns on their use at home.

What do you do to shut out the world?







Phones!

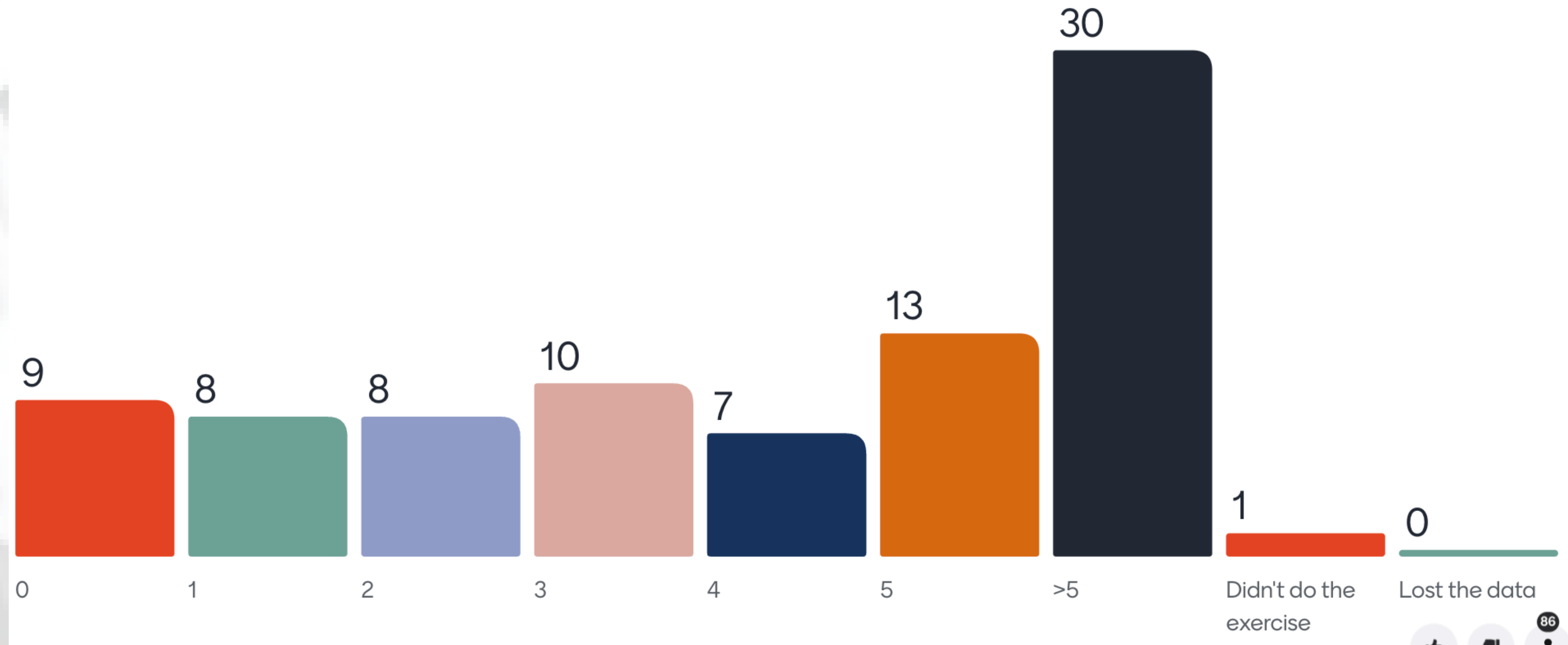


Distraction study

Notifications off

4 minutes of silent reading

How many times were you distracted?



How many minutes was it until your *first* distraction?

65 responses



noises

What distracted you?

people whispering in the background and leaving the room

People talking, people standing up

people laughing, pens tapping

can't remember

dropped water bottle

the thought of pressing lap

People were leaving and I was sitting at the back :((

tapping pencil, changing lec slides, coughs and people leaving

Someone getting up to leave the room

chatters.

Phone

movements on the projected screen

phone

Sounds around me

What distracted you?

someone tapping on the desk

People whispering

A notification on my phone

screen

My own thoughts

Environment

noise

Tapping

Noises around me

dunno

Finshed

Myself

People talking and thoughts of other people

people talking in the middle, someone coughing

What distracted you?

person near me , then slides moving on screen later on too	Thoughts sleepy sounds
Sound	Sounds around me, changing of the projector screen
Thoughts	Sound
people whispering	A random song in my head
Nothing	

2023



Distractions, categorised

Sensory events: noise, other peoples laptops, dense slides, muscle soreness, *phone*

Life issues: issues in my life, other works, other deadlines

Drives: hunger, sleepiness, boys

Cognitive, interest, learning: boring lectures, social media - instagram, *phone*, tangents, spring break planning, content is too hard or ez

HARD QUIZZ



HARD QUIZZ



Distractions, categorised

Sensory events (bottom-up): noise, other peoples laptops, dense slides, muscle soreness, *phone*

Life issues (top-down): issues in my life, other works, other deadlines

Drives: hunger, sleepiness, boys

Cognitive, interest, learning (top-down): boring lectures, social media - instagram, *phone*, tangents, spring break planning, content is too hard or ez

Distractions, categorised

Sensory events (bottom-up): noise, other peoples laptops, dense slides, muscle soreness, *phone* [Chapter 5](#)

Life issues (top-down): issues in my life, other works, other deadlines. NOT COVERED IN PSYC2016

Drives: hunger, sleepiness, boys. NOT COVERED IN PSYC2016.

Cognitive, interest, learning (top-down): boring lectures, social media - instagram, *phone*, tangents, spring break planning, content is too hard or ez. MULTIPLE CHAPTERS MENTION MEMORY, + [CHAPTER 16](#)

Learning

Association of stimuli with reward ([Chapter 16](#))

- When you open an app, you get interesting things
- Not just interesting, but also a social reward (everyone wants to be liked)

What distracts you during studying?

phone; phone; Phone; Phone; phone; Phone; phone; my phone; cellphone; My phone; buzzing phone; Phone buzzing; My damn phonnneeee :(

My phone, the habit of picking it up all the time
messages/instagram

flat mate coming into my room

The clock

Social media

my boyfriend playing video games right next to me

snacks

vacuumming in the background, remembering random things

My own thoughts

Cans of kombucha being drunk.

my housemates

Social facilitation

Shut Up
& Write!®

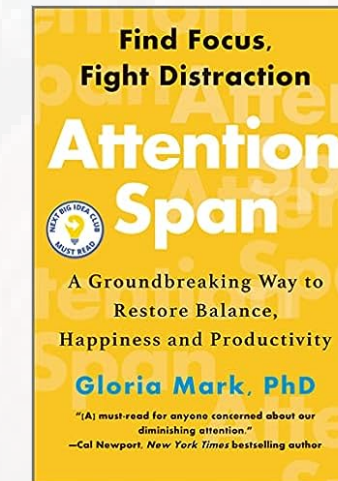
Every Tuesday from
4:00 PM – 5:30 PM

- Harness social attention to keep you on task! - Study groups - Group workouts

In 2004, the average time people spent on a single screen was 2.5 minutes.
“I was astounded. That was so much worse than I’d thought it would be,” said Gloria Mark.

But that was just the beginning.

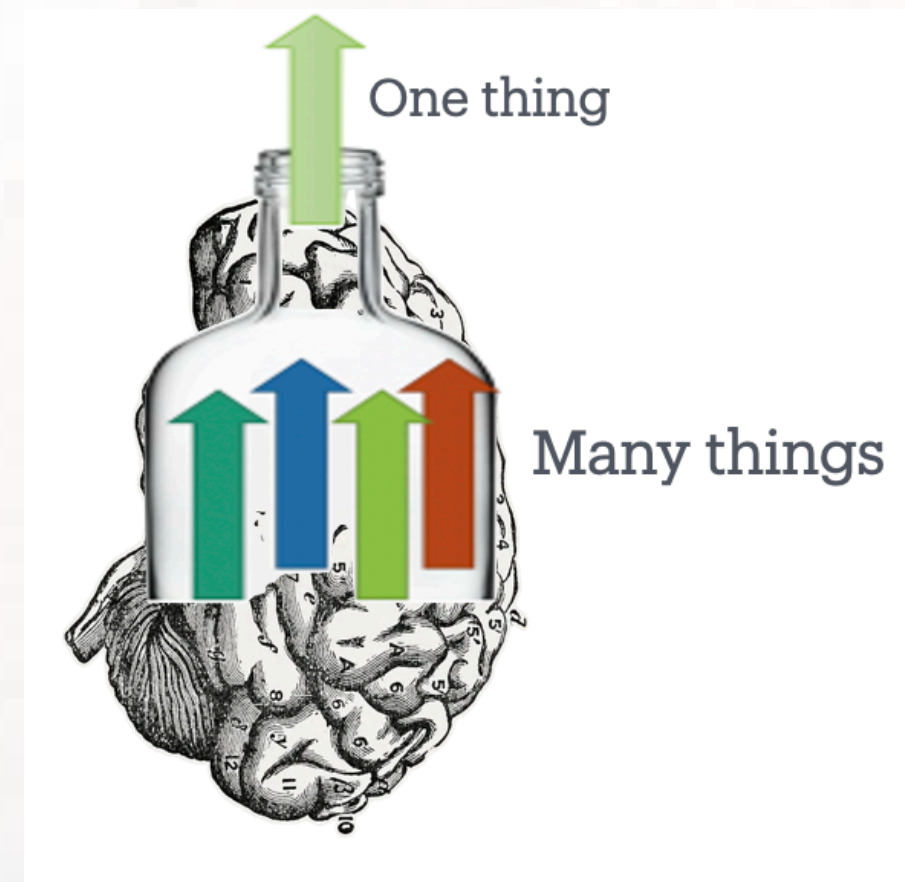
By 2012, Mark and her colleagues found the average time on a single task was 75 seconds. Now it’s down to about 47.



Multi-tasking?



- We can only focus on one thing at a time.



- Task-switching.
 - “It’s like we have an internal whiteboard in our minds,” Mark said.
 - “If I’m working on one task, I have all the info I need on that mental whiteboard. Then I switch to email. I have to mentally erase that whiteboard and write all the information I need to do email. And just like on a real whiteboard, there can be a residue in our minds. We may still be thinking of something from three tasks ago.”

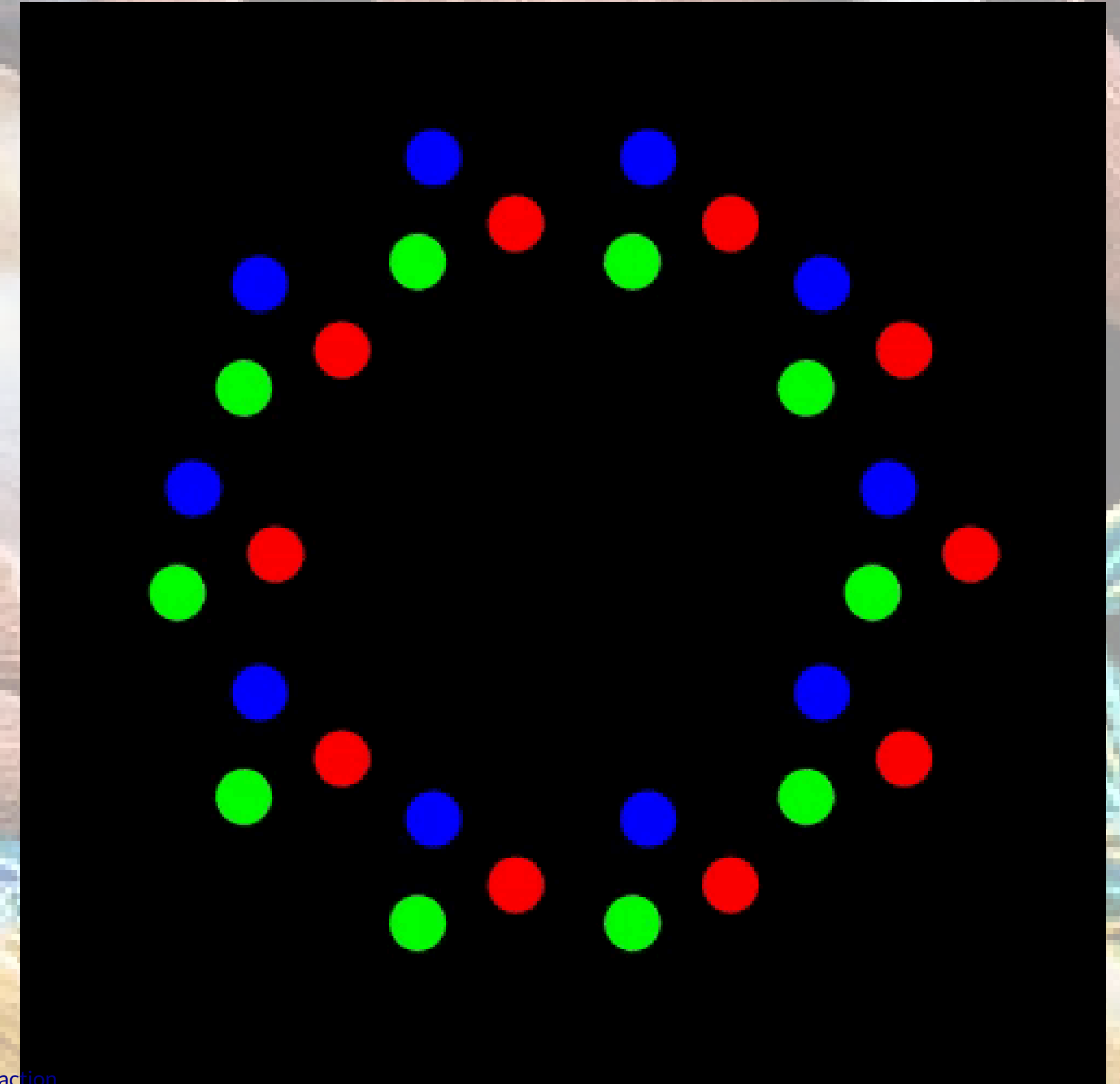


When have you been most focused?

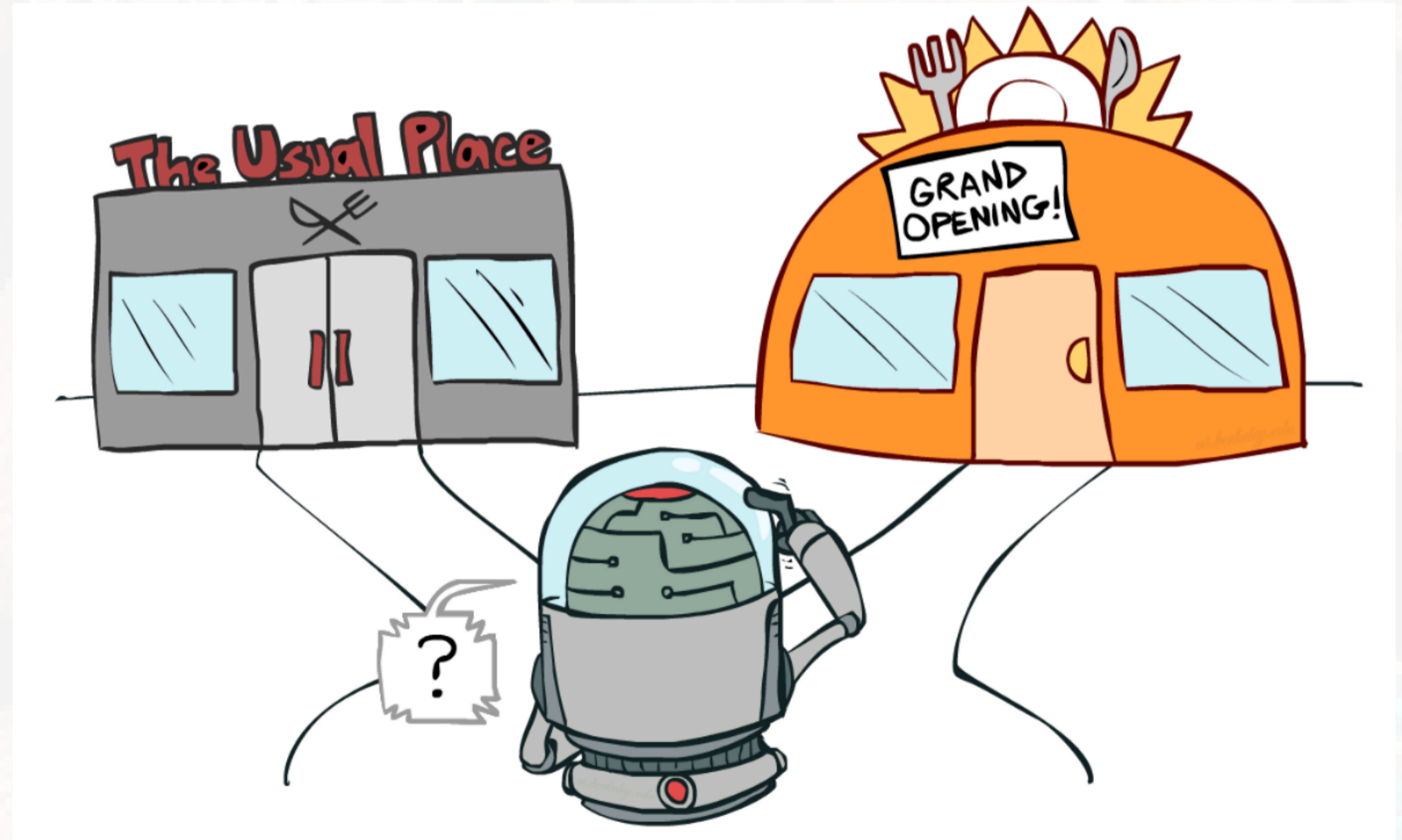
Menti.com

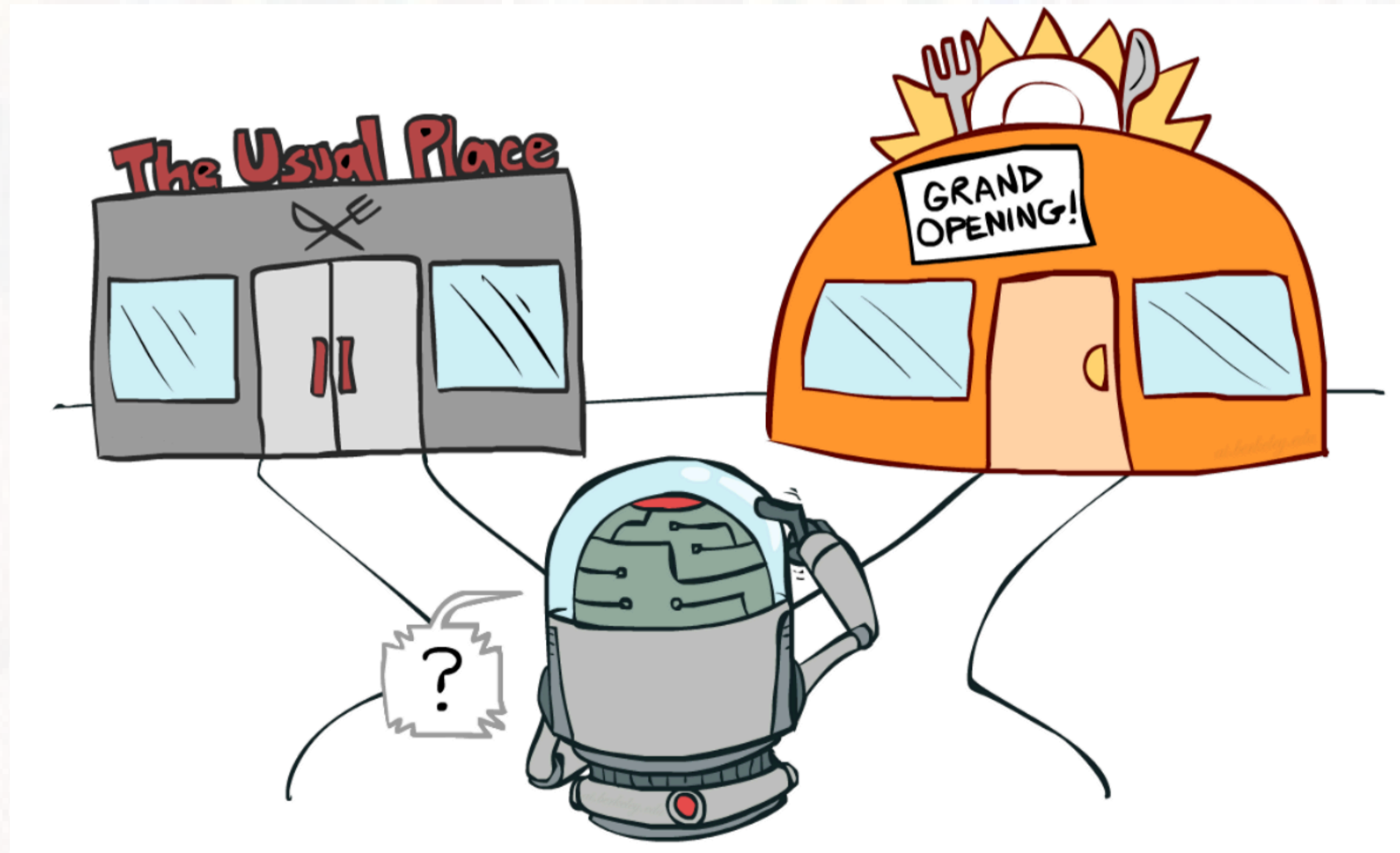
Get involved in research!

alex.holcombe@sydney.edu.au



- When we find something rewarding/worthwhile, ideally we **exploit** it
- But exploration is also a drive
 - The *yin* to exploitation's *yang*
 - The **explore/exploit trade-off**
- Vigilance is more about threats than about exploration





Exploitation - using known information to get rewards

Exploration - exploring the environment to look for new information/rewards

All day, the world makes its demands. There's so much of it, world
begging to be noticed.
ultrawhelmed sensorium, my self





ultrawhelmed sensorium, my self
taxed with being a self, brimming with living's rowdy mechanics
and disruptions unremitting,
a thought flits by, then another (an unpaid bill, a jingle's tenacious refrain)
and, ... I attend, as best I can,...
yielding to the most persuasive god, the most recent,
to each thing I say Yes?
Yes!
zealot of whatever calls me next.

menti.com

Thing(s) I didn't
understand

Waiting for responses ...



menti.com

Thing(s) I didn't understand

Waiting for responses ...

